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Duck Eats Rabbit: Exactly Which Type of Relational Phrase Can Disambiguate the Perception of Identical Side by Side Ambiguous Figures?

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Abstract
Many individuals cannot at first see two ambiguous figures as different interpretations simultaneously, even with effort. Here in a large sample replication, we find that the phrase "duck eats rabbit" allows those who could not see a duck and rabbit side by side to do so. In a second experiment, we show that a relational phrase "next to" that does not disambiguate the spatial position interpretation does not similarly allow the duck to be seen next to the rabbit, supporting the proposal that top-down semantic-framing can influence perception of ambiguous figures.

Keywords
ambiguous figure, illusion, endogenous/exogenous, semantics, top-down

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In this article, we show evidence that top-down influences of semantic framing can bias the perception of different interpretations of two identical side-by-side ambiguous figures (Toppino & Long, 2015). In previous work (Jensen & Mathewson, 2011), around 40% of respondents were not able to simultaneously perceive the duck interpretation next to a rabbit interpretation, even when asked to "try hard." Asking *non-seers* to imagine that the "duck is eating the rabbit" induced an immediate flip in interpretation; subjects previously unable to see two interpretations at once, side by side, were now able to.

Here, we replicate this effect, and in a second experiment, test the specificity of the relational phrase needed. We ran identical experiments in two semesters, with a single

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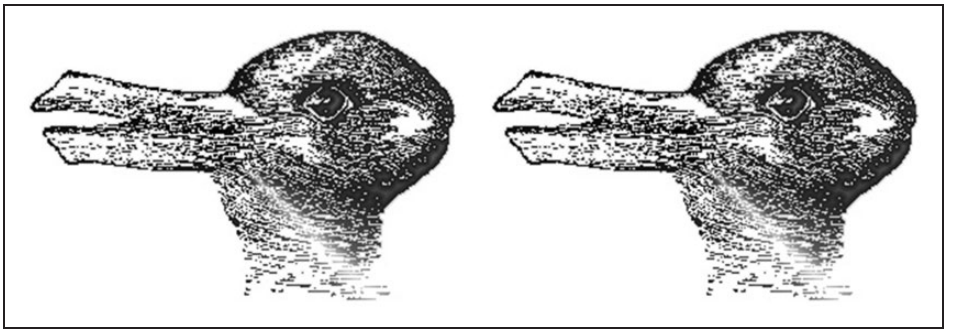


Figure 1. Pair of ambiguous figures used in the experiment adapted from Jastrow (1899).

change to the wording of the key phrase. In Experiment 1, we replicated Jensen and Mathewson (2011) and told participants to, ''imagine the duck is eating the rabbit.'' In Experiment 2, participants were asked to, ''imagine the duck is next to the rabbit.'' This phrase has similar spatial components without disambiguating the positions of the two animals. This control also allows us to rule out experimental bias in the questions as these are held constant.

Participants answered the questions in a web interface, as part of a mass-testing survey for the introductory psychology research subject pool at the University of Alberta, Department of Psychology 2015–2016. Participants received credit for completion, gave informed consent prior to participation, and the survey was approved by the internal Research Ethics Board of the University.

In Experiment 1, participants answered five questions, while seeing the images in Figure 1 with the phrase ''Use the following image to answer the questions below.'' First participants were shown a single duck/rabbit, and asked (Q1) ''What kind of animal did you see first?'', selecting either duck or rabbit. After advancing, a pair of identical stimuli were presented and participants were asked: (Q2) ''If you can make sense of both figures at once, what did you see first?'', (Q3) ''Can you see both images as the same figure'' (Yes vs. No), and (Q4) ''Can you see them as different figures at the same time? (Yes vs. No). After another button press, the imperative phrase was presented ''Imagine the duck is eating the rabbit,'' with the image and (Q5) ''Now can you see them as different figures at the same time?'' (Yes vs. No).

Of the 1,971 respondents in Experiment 1, a large proportion could see matched interpretations when asked (Q3; 1,857/1,963 responses; 94.6%). Only a small proportion of subjects were coded as indicating, uncued, that they spontaneously saw a duck next to a rabbit (Q2; 43/1,878; 2.3%; Figure 2). When explicitly asked to try to see unmatched interpretations, the majority of participants were able to, which we refer to as *seers* (Q4; 1,213/1,959; 61.9%). The 746 remaining respondents, non-seers, were unable to see a duck beside a rabbit.

Providing the imperative phrase ''duck eating the rabbit'' increased the overall number of respondents who indicated they could see a duck beside a rabbit (Q5; 1,687/1,949; 86.6%; $\chi^2(1) = 308.56$, $p < .0001$). Of the non-seers from Q4 who went on to respond to Q5, many were now able to see a duck beside a rabbit (Q5|∼Q4; 580/739; 78.5%).

In Experiment 2, participants answered the same five questions, while seeing the same images. All questions were identical in Experiment 2 except the imperative phrase which was now ''Imagine the duck is next to the rabbit''. A total of 1,652 students answered this new

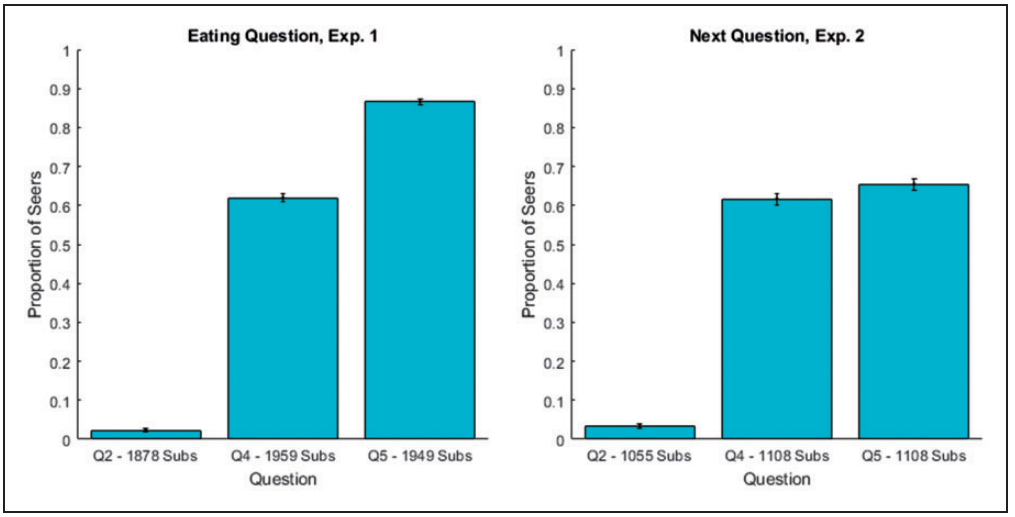


Figure 2. Proportion of individuals indicating they saw a duck and rabbit side by side following each imperative question, for both experiments. Error bars represent standard error of the mean.

series of questions. Of these, 532 had already completed Experiment 1 and their data were removed from all further analyses. A large proportion of respondents could see matched interpretations when asked (Q3; 1,063/1,110 responses; 95.8%). Only a small proportion of subjects were coded as indicating, that they spontaneously saw a duck next to a rabbit (Q2; 35/1,055; 3.3%). When explicitly asked to try, again there were a large proportion of seers (Q4; 683/1,108; 61.6%). The 425 remaining non-seers were unable to see the duck beside the rabbit.

Providing the control phrase, "duck next to the rabbit" increased the overall number of respondents who indicated they could see unmatched interpretations, but much less than in Experiment 1 (Q5; 725/1,108; 65.4%; $\chi^2(1) = 3.27$, $p = .07$). Of the non-seers from Q4 who went on to respond to Q5, less than half were now able to see the unmatched interpretations (Q5|~Q4; 178/421; 42.3%), much less of an increase than in Experiment 1 ($\chi^2(1) = 153.65$, $p < .0001$).

The results confirm that relational phrases can allow perception of two different ambiguous interpretations side by side. The large-sample replication supports the reliability of this finding, and the new experimental interface and recruitment (online course credit vs. paper and pencil campus recruitment) indicate generalizability. The specific phrase "duck eating the rabbit" disambiguates the two figures. Saying the duck and rabbit were "next to" one another did not help individuals see the duck and rabbit side by side. Since both indicate spatial relations, an explanation of the effect as a change from local to global processing of the two figures is ruled out (Navon, 1977). These results provide tentative evidence that top-down semantic framing of the relationship between the two figure interpretations may bias perception of the ambiguity (see Firestone & Scholl, 2016). However, sources of biasing at a range of levels from changing eye movements and fixations to boosting of high-level representation or biasing response preparation should be further considered to determine whether the effect is truly top-down. Finally, objective psychophysics tests of this linguistic biasing of perception are needed to better understand how a small semantic cue can at last let ducks and rabbits stand side by side.

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